Predicting the performance of similarity index via leave-one-out method
—Model study and empirical tests—

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Abstract

Similarity index, measuring structural proximity of node pairs, is used to find similar nodes in networks. The performance of similarity index depends on the topological property of a given network, so it is important to find which similarity shows best performance in a given network. However, performance of index cannot be measured without additional information of nodes. In this presentation, we employ leave-one-out method to predict the performance of the index without additional information. We applied similarity index to stratified model network, and AUC value is used for quantifying the performance. We observed the linear correlation between AUC value from the node age information and AUC value from leave-one-out method.

Keyword: Similarity index, stratified model network, leave-one-out method, AUC value

Similarity index is the measure that quantifies topological proximity between nodes. Previous studies focused on investigation of indices that shows best performance in the given system. However, performance can be measured with additional information of nodes, so it cannot be measured without that. In this presentation, we investigate that leave-one-out method [1] gives some clue for the prediction of performance of given similarity for given network.

Stratified model network [2] is used for model study. Each index has intrinsic parameter called 'age', given as random integer between 1 and 10. Connection probability is given as $P_0 e^{-a\Delta t}$, where P_0 and a is constant and Δt is age difference between two nodes.

We employ AUC value to quantify the result. This problem can be transformed into binary classification problem, where node pairs are classified into true if their ages are the same and false if not.

To predict the AUC value, we apply leave-one-out method. It calculates AUC value for one missing link via similarity index. 10 indices are employed.

As a result, the AUC value from age information and AUC value from leave-one-out method are linearly correlated. This result implies that we can predict the performance of similarity index only using topological information.

We will discuss about the result of empirical tests and the reason why the linear correlation holds.

References

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